a.Cohort and EHR phenotype

All of Us WGS + EHR
295,706 female at birth

→ All of Us ancestry prediction
EUR label · max P > 0.75
150,900

→ Phenotype + unrelated/QC
143,255

→ **EHR cancer phenotype**
Case: breast cancer EHR **8,528**
Control: no malignancy EHR **134,727**
Other malignancy, no BC: excluded

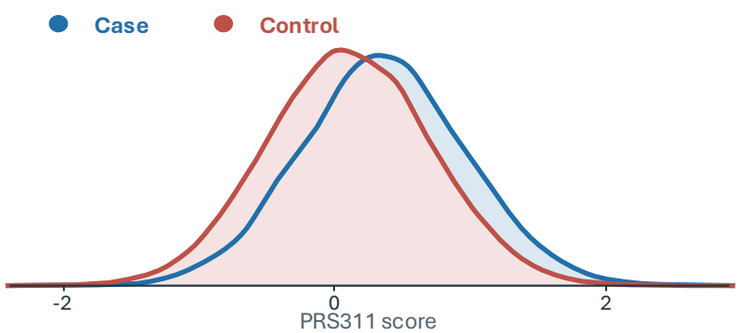
b.Common variants to PRS score

genotypes

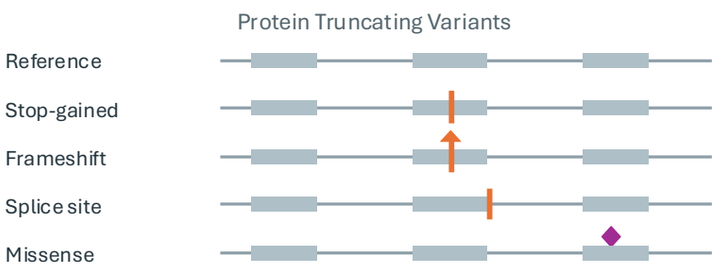
SNP 1	SNP 2	SNP 3
dosage = 0	dosage = 1	dosage = 2
× w1	× w2	× w3

filled circle = effect allele

$$\mathrm{PRS311}_i = \sum_{j \in J_{311}} w_j \, d_{ij}, |J_{311}| = 302$$



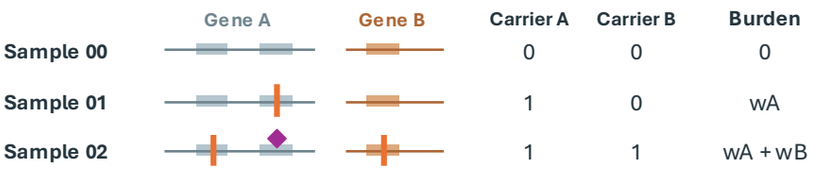
c. Rare variants Burden$_{rare}$



PTV and LOFTEE HC + P/LP and ClinVar ≥2 stars

variants passed QC, GQ ≥20, and rare AF <1%

gene carrier status



$$\mathrm{Burden}_i^{\mathrm{rare}} = \sum_{g \in G} C_{ig} \ln(\mathrm{OR}_g), |G| = 13$$

PRS + Burden$_{rare}$

d. Downstream analyses

Case-Control: n=141,204
8,528 cases, 132,676 controls

ID	PRS	Status
00001	-1.2	case
00002	-0.4	control
00003	+1.7	control

Variant	SNP 1	SNP 2	SNP 3
Effect Allele	1	0	2
Weight	0.10	-0.04	0.20

scores matrix → logistic → CV 5-fold × 10 → AUC

$$\mathrm{logit}\{\Pr(Y_i = 1)\} = \beta_0 + \beta_P \mathrm{PRS}_i + \beta_B B_i + \beta_{PB}(\mathrm{PRS}_i B_i) + \boldsymbol{\gamma}^{\mathrm{T}} \mathbf{X}_i$$

covariates include: last EHR age, PCs 1–10, centre, sequencer

134,727 - excluded 2,051 controls: last EHR age unavailable

Age at diagnosis: cases-only
7,996 analyzed cases

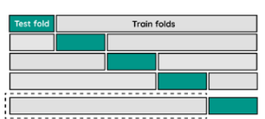
ID	quanlify gene	burden
00001	None	0
00002	CHEK2	ln(2.54)
00003	BRCA2	ln(10.57)
00004	ATM,BRCA2	ln(1.35)+ln(10.57)

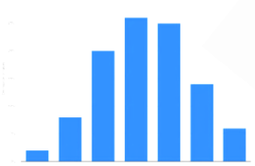
scores matrix → diagnosis age → OLS → β estimate

$$A_i = \alpha_0 + \alpha_P P_i + \alpha_B B_i + \alpha_{PB}(P_i \times B_i) + \sum_{k=1}^{10} \delta_k \, \mathrm{PC}_{ki} + \varepsilon_i$$

covariates include PCs 1–10; HC1 robust standard errors

8,528 - excluded 532: breast cancer record = first EHR record







ID	quanlify gene	burden
00001	None	0
00002	CHEK2	ln(2.54)
00003	BRCA2	ln(10.57)
00004	ATM,BRCA2	ln(1.35)+ln(10.57)

Variant	SNP 1	SNP 2	SNP 3
Effect Allele	1	0	2
Weight	0.10	-0.04	0.20

ID	PRS	Status
00001	-1.2	case
00002	-0.4	control
00003	+1.7	control